

Balancing

Reference: Applied Mechanics: Hannah & Hillier and Reed's Vol. 2

Newton's Laws of Motion

First Law — Inertia

A body remains at rest or in uniform straight-line motion unless acted upon by a net external force.

A body in motion travels in a straight line unless forced to change direction. This resistance to any change in motion is called **inertia**. The law defines what a force *is*: the only thing capable of altering a body's state of motion.

$$\sum \vec{F} = 0 \implies \vec{v} = \text{constant}$$

Second Law — Force and Acceleration

The net force acting on a body equals its mass multiplied by its acceleration.

$$\boxed{F = ma}$$

Force and acceleration act in the same direction. The relationship is linear: doubling the net force doubles the acceleration; doubling the mass halves it for the same force.

Change	Effect on acceleration
Force doubled	Acceleration doubled
Mass doubled	Acceleration halved
Both doubled	Acceleration unchanged

Third Law — Action and Reaction

For every force one body exerts on another, the second body exerts an equal and opposite force back on the first.

$$\vec{F}_{A \rightarrow B} = -\vec{F}_{B \rightarrow A}$$

- The forces are **equal in magnitude** and **opposite in direction**.
- The reaction force has the same nature as the action (both contact, both gravitational, etc.).

Centrifugal and Centripetal Forces

A body in motion travels in a straight line unless forced to change direction (Newton's First Law). When moving in a circle, its inertia causes it to resist that continuous change of direction — in a rotating reference frame, this resistance appears as a fictitious outward “centrifugal” force. The real inward force that continuously deflects the body onto the circular path is the **centripetal force**. Because circular motion involves a constant change of direction, velocity is always changing even at constant speed. For small angles, the magnitude of this change in velocity simplifies to $v\theta$.

Static Balance

A system is statically balanced when the resultant of all centrifugal forces due to rotating masses is zero — i.e., the centre of mass lies on the axis of rotation. Static balance requires the vector sum of all mr terms to be zero:

$$\sum mr = 0$$

A single correction mass placed in the same plane can achieve static balance.

Dynamic Balance

Static balance alone is insufficient for systems with masses distributed across different axial planes. Dynamic balance requires two conditions to be satisfied simultaneously:

1. The vector sum of all centrifugal forces equals zero (static condition).
2. The vector sum of all **moments** of these forces about any reference plane equals zero.

$$\sum mr = 0 \quad \text{and} \quad \sum mr\ell = 0$$

This generally requires **two correction masses** placed in two chosen planes.

Balancing in a Single Plane

For coplanar rotating masses, a graphical or analytical approach uses the **force polygon**. Closing the mr vector polygon determines the required balancing mass and its angular position.

Balancing in Several Planes

For masses distributed along a shaft, the procedure is:

1. Choose a reference plane and construct a **couple polygon** using mrl terms, where ℓ is the axial distance from the reference plane.
2. Closing the couple polygon gives the first correction mass (magnitude and angle).
3. Construct the **force polygon** using mr terms to find the second correction mass.

Balancing of Reciprocating Machinery

A reciprocating engine generates forces that are fundamentally harder to balance than rotating masses. The piston moves in one plane only, the problem cannot be solved by a single counterweight alone.

Primary and Secondary Forces

Treating the connecting rod as two lumped masses — one at the crank pin, one at the gudgeon pin — splits the system into a rotating part and a pure reciprocating part. The total reciprocating force is:

$$F = mr\omega^2 \left(\cos \theta + \frac{r}{\ell} \cos 2\theta + \dots \right)$$

Term	Name
$\cos \theta$	Primary force
$\frac{r}{\ell} \cos 2\theta$	Secondary force
Higher terms	Negligible in practice

The secondary force arises from the obliquity of the connecting rod; higher harmonics are generally ignored. Written separately:

Primary force (first-order):

$$F_{\text{primary}} = mr\omega^2 \cos \theta$$

Secondary force (second-order):

$$F_{\text{secondary}} = \frac{mr\omega^2}{n} \cos 2\theta$$

where $n = \ell/r$ is the ratio of connecting rod length ℓ to crank radius r .

Methods for Complete Balancing

Crankshaft Counterweight (Partial Balance)

Adding a mass opposite the crank pin cancels the rotating component entirely but only reduces the primary reciprocating force — it cannot eliminate it. This trades a vertical unbalance for a smaller horizontal one.

Multi-Cylinder In-Line Engines

By phasing cranks at equal angles, primary and secondary forces and couples can cancel across the engine. Balance requires:

$$\sum mr = 0 \quad \text{and} \quad \sum mrl = 0$$

for both primary and secondary components.

In multi-cylinder engine balancing there are four things to check:

- **F (Force)**: the vector sum of all mr terms. If non-zero, there is a net radial force shaking the engine.
- **C (Couple)** — the vector sum of all mrl terms. If non-zero, there is a rocking moment twisting the engine about a transverse axis, even if the net force is zero.

Configuration	Primary F	Primary C	Secondary F	Secondary C
2-cyl, 180°	✓	✗	✓	✓
4-cyl, 180°	✓	✓	✓	✗
6-cyl, 60°	✓	✓	✓	✓

The **six-cylinder in-line engine** achieves complete balance for both primary and secondary effects.

Partial Balance

Complete balance is not always possible or practical — particularly for single-cylinder engines. In such cases a balance mass is added to reduce the dominant unbalanced force, and a residual force in one direction is accepted to eliminate the more harmful force in the perpendicular direction.

Critical Speed

Every rotating shaft is also an elastic beam with natural frequencies of lateral vibration. The **critical speed** is the rotational speed at which the excitation frequency from residual unbalance equals a natural frequency, producing resonance.

For a shaft carrying a disc, using the **static deflection** δ_{st} under its own load:

$$N_c = \frac{60}{2\pi} \sqrt{\frac{g}{\delta_{st}}}$$

This is convenient because δ_{st} can be directly measured or calculated from beam theory.

Behaviour Around Critical Speed

Speed regime	Shaft behaviour
Well below N_c	Shaft runs stiffly; centre of mass geometric centre
Approaching N_c	Amplitude grows rapidly — whirling begins
At N_c	Resonance; amplitude theoretically unbounded
Well above N_c	Shaft self-centres; centre of mass migrates toward axis

Many turbines and high-speed spindles operate **supercritically**, above the first critical speed, and simply pass through it during run-up.

Multiple Critical Speeds

A real shaft has infinitely many lateral modes, each giving a critical speed:

$$N_{c_n} \propto n^2 \quad (\text{uniform simply-supported shaft})$$

In N_{c_n} , the subscript n is the **mode number** — an integer (1, 2, 3, ...) identifying which lateral vibration mode is being referred to.

- N_{c_1} — **first critical speed**, corresponding to the fundamental (lowest) mode shape, where the shaft bows into a single half-wave.
- N_{c_2} — **second critical speed**, higher in frequency, less likely to be encountered in normal operation.
- N_{c_3} — **third critical speed**, and so on.

The relationship $N_{c_n} \propto n^2$ means the second critical speed is roughly 4× the first, and the third is roughly 9× the first. In practice, only N_{c_1} is of primary design concern.

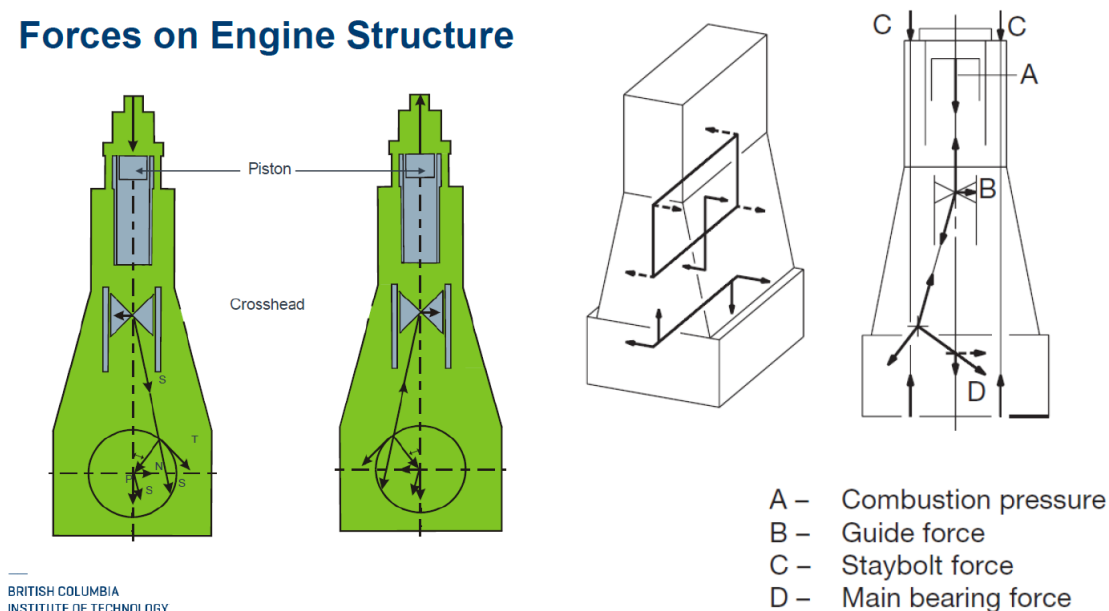
The **first critical speed** is the most dangerous: it is encountered first and produces the largest deflection amplitude.

Vibration

Any elastically-coupled shaft, or other systems, will have one or more natural frequencies, which, if excited, can build up to an amplitude which is perfectly capable of breaking crankshafts. 'Elastic' in this sense means that displacement or a twist from rest creates a force or torque tending to return the system to its position of rest, and which is proportional to the displacement. An elastic system, once set in motion in this way, will go on swinging, or vibrating, about its equilibrium position forever, in the theoretical absence of any damping influence. The resulting time/amplitude curve is exactly represented by a sine wave; that is, it is sinusoidal.

Ship machinery installations have two principal sources of excitation: the main engine(s) and the propeller(s). The two components are essentially linked by elastic shaft systems, and may also include gearboxes and elastic couplings. The whole system is supported in flexible hull structures, and the forms of vibration possible are, therefore, diverse. The greater complexity of vibration problems dictates a larger number of calculations to ensure satisfactory vibration levels from projected installations. For optimum results, the vibration performance of the plant has to be investigated for all anticipated operational modes.

Forces on Engine Structure



Critical Speed

The critical speed of an engine is the rotational speed at which the engine's components begin to vibrate in an unstable manner. There are a number of forces acting on an engine from each cylinder, and these can vary in magnitude and direction as the engine rotates. The turning moment acting on the crankshaft, and the vibration effect it produces, is complex, and repeated at every revolution. Like any other repeating curve, it can be represented by a series of sine waves, each with different amplitudes and phases to the basic underlying sine wave. Each of the sine waves, or harmonics, can excite the free natural frequency of the engine system, and the speeds at which the excitation takes place are called the critical speeds.

There are many critical speeds, but most are outside of the normal operating speed of the engine. The critical speeds are distinguished from each other by the number of vibrations that occur during each engine crankshaft revolution. At the first critical speed, there is one vibration at each crankshaft revolution, and this is known as the first order vibration. At the second critical speed, there are two vibrations during each revolution of the crankshaft, and this is the second order vibration. For a fourth order vibration, there would be four vibrations during each revolution of the crankshaft. The orders of vibration that are significant for a particular crankshaft, and are likely to produce large vibration amplitudes, depend on the number of engine cylinders. The important orders are usually a multiple of the number of cylinders, so for a four-cylinder engine, the 4th, 8th, 12th, and 16th orders of vibration will be significant, and likely to present vibration problems if they occur in the engine operating speed range. For a nine-cylinder engine, the 9th and 18th orders of vibration will be significant, although there is less likelihood of the engine operating at a speed where the 9th order vibration is likely to occur. The firing order of the cylinders is influential in reducing the crankshaft vibration.

Summary of Equations

Concept	Equation
Centrifugal force	$F = mr\omega^2$
Static balance	$\sum mr = 0$
Dynamic balance	$\sum mr = 0$ and $\sum mr\ell = 0$
Primary reciprocating force	$F_{\text{primary}} = mr\omega^2 \cos \theta$
Secondary reciprocating force	$F_{\text{secondary}} = \frac{mr\omega^2}{n} \cos 2\theta$
Critical speed	$N_c = \frac{60}{2\pi} \sqrt{\frac{g}{\delta_{st}}}$

Summary

Balancing is fundamentally a **vector problem**. Both graphical (polygon) and analytical methods are used throughout. Dynamic balancing always requires satisfying both force *and* moment equilibrium, building progressively from simple single-plane cases up to the complexity of multi-cylinder engine analysis. Critical speed analysis adds a further dimension: even a perfectly balanced shaft can experience destructive resonance if its operating speed coincides with a lateral natural frequency.